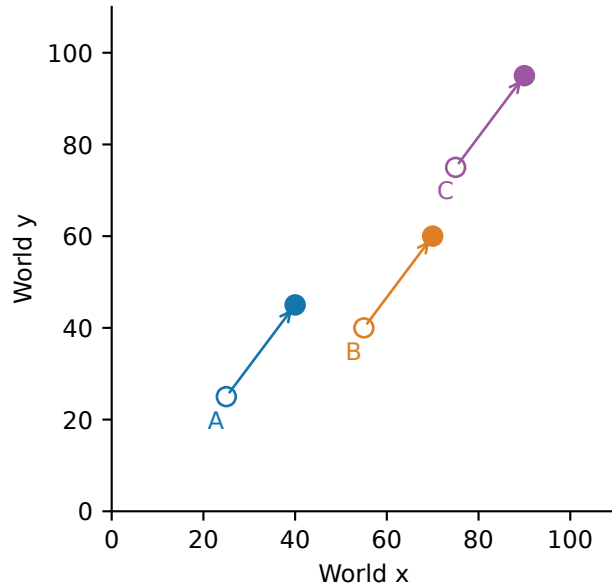
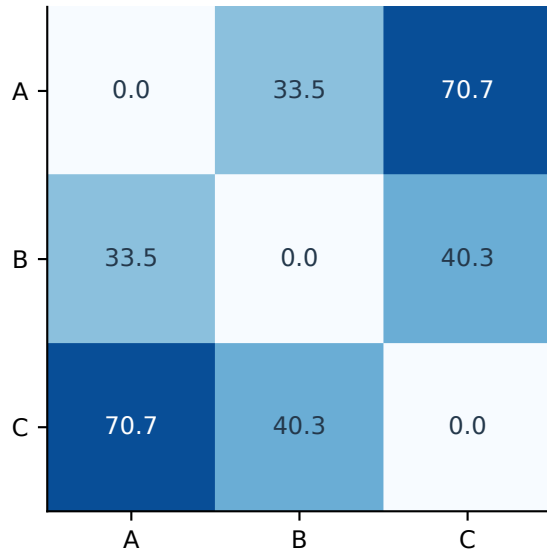


Relative distances are translation-invariant; fixed references retain the missing signal

(a) Common translation $v = (15, 20)$



(b) ST-MTM distance input unchanged



$D(C + v) = D(C)$
Distances alone cannot identify translation

(c) Dual reference changes on both axes

